

Advanced JavaScript Frameworks (Angular)

CIA-2

- **Project Title:** *Responsive Portfolio Website Detailing Movie Ticket Booking and Theatre Management System.*
- **Submitted By:**
 - 2462306 Adithyan CR
adithyan.cr@btech.christuniversity.in
 - 2462308 Adwaith Praveen
adwaith.praveen@btech.christuniversity.in
 - 2462331 Jerolin Mathew
jerolin.mathew@btech.christuniversity.in
 - 2462332 Joel Anthony Dsilva
joel.anthony@btech.christuniversity.in
- **Course:** *Advanced JavaScript Frameworks (Angular)*
- **Instructor Name:** Mr. Narendra
- **Institution:** *Christ University*
- **Date of Submission:** 23/01/2026
- **GitHub Repo:**
<https://github.com/joel-dsilva/-Movie-Ticket-Booking-and-Theatre-Management-System>
- **Published Output:**
<https://joel-dsilva.github.io/-Movie-Ticket-Booking-and-Theatre-Management-System/>

2. Abstract

This project presents the design and implementation of a Movie Ticket Booking and Theatre Management System using Angular and TypeScript. The application allows users to browse



movies, view showtimes, select theatres, choose seats, and confirm bookings through a smooth, single-page application (SPA) experience. The system is built using Angular's component-driven architecture, routing, services, dependency injection, reactive forms, and observables to ensure modularity and maintainability. The project also emphasizes a responsive UI using Angular Material, providing user-friendly navigation and dynamic seat selection. Overall, the application demonstrates real-world booking workflows and strengthens practical understanding of Angular framework concepts.

3. Objectives

- Design a user-friendly and responsive restaurant ordering application
- Implement Angular components for modular development
- Use Angular routing for navigation between Movies, Theatres, and Bookings
- Enable viewing of showtimes and performing seat selection with booking confirmation
- Use Angular services to manage booking state and theatre/movie data
- Apply modern UI/UX principles suitable for real-world applications

4. Scope of the Project

- Front-end web application developed using Angular
- Includes browsing movies, viewing showtimes, seat selection, and booking confirmation
- Uses static JSON files or a mock backend (JSON Server) for movies, theatres, showtimes, and bookings
- No real payment gateway integration in the current version
- Designed for desktop and responsive layouts
- Intended for academic and learning purposes

5. Tools & Technologies Used

Tool/Technology	Purpose
HTML5	Markup and content structure
CSS3	Styling and layout management
Angular	Front-end framework
Typescript	Application logic
VS Code	Code editor
Angular CLI	Project scaffolding and build

Node.js	Runtime environment required for Angular
Chrome DevTools	Testing and debugging

6. Application Structure Overview

- Component-based architecture (Home, Menu, Cart, Checkout)
- Reusable UI components for movie cards, showtime cards, and seat grid
- Angular services used for movie/theatre data retrieval and booking state management
- Routing module for seamless navigation
- Separation of concerns between UI and logic

7. Styling and UI Strategy

- Component-level CSS for styling
- Card-based layout for displaying movies and theatres
- Consistent theme, spacing, and typography across pages
- Responsive layout using Angular Material grid and flexible containers
- Clear visual seat selection states (Available / Selected / Booked)
- User feedback through alerts and buttons

8. Key Features

Feature	Description
Movie List	Displays currently showing movies
Movie Detail	Shows movie description, duration, cast, rating, and poster
Showtime List	Displays available showtimes for selected movie and theatre
Theatre Selection	Allows users to choose theatre location/screen
Seat Selection	Interactive seat grid to select and reserve seats
Booking Summary	Displays booking confirmation details (movie, theatre, time, seats)
Form Validation	Validates user details (name/email/phone) during booking

9. Challenges Faced & Solutions

Challenge	Solution
Managing booking flow across multiple pages	Implemented Angular service
Page navigation handling	Used Angular Router
Maintaining UI consistency	Applied reusable components
Seat selection UI complexity	Built a seat-grid component using *ngFor and state flags
Maintaining selected seats during navigation	Stored selection in service/state and optionally LocalStorage
Handling dynamic showtime/movie data	Used mock JSON + observables for async binding

10. Outcome

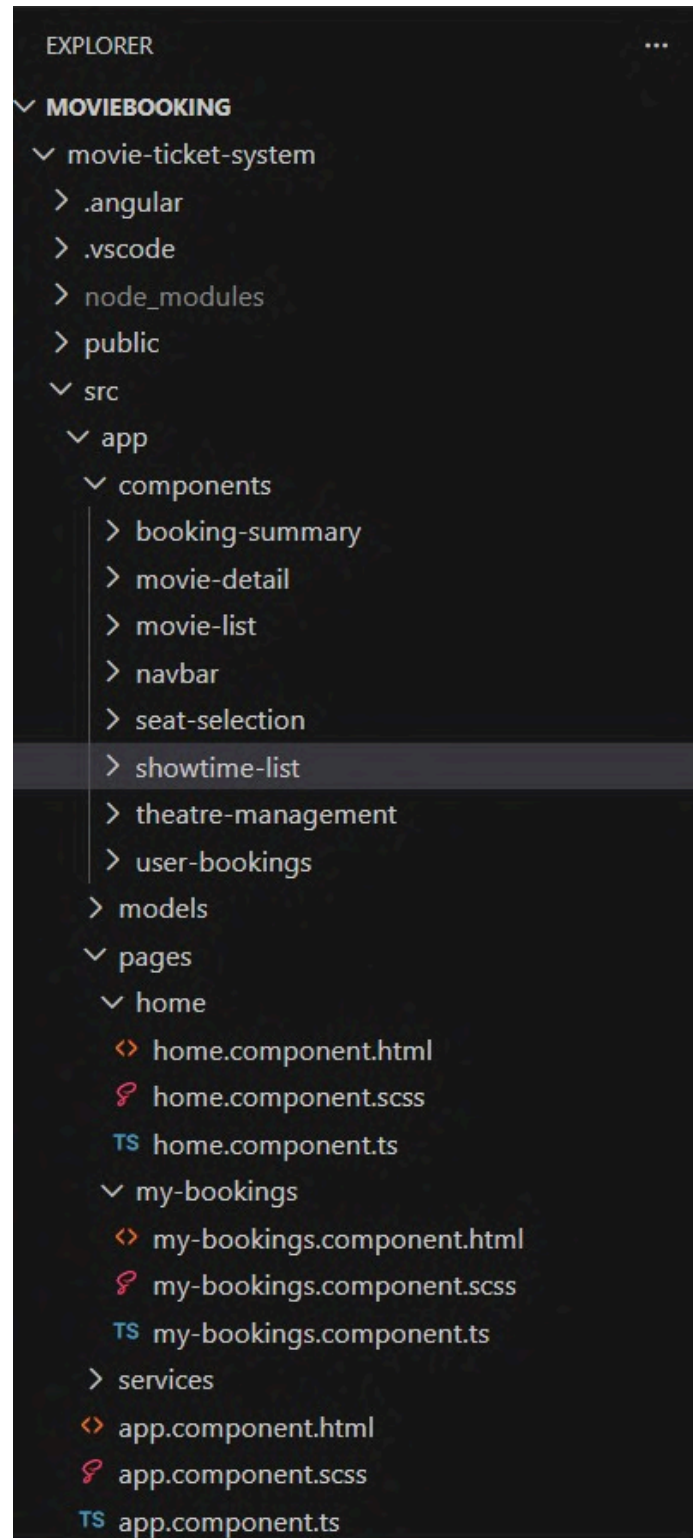
- Successfully developed a functional Movie Ticket Booking and Theatre Management SPA
- Implemented key Angular concepts: components, routing, services, DI, forms, and observables
- Achieved a clean, responsive, and user-friendly UI suitable for real-world booking workflow
- Strengthened understanding of modular Angular architecture and state management

11. Future Enhancements

- Backend integration with database support
- Payment gateway integration
- Admin panel for theatre management (add screens, showtimes, pricing)
- User authentication and order history
- Real-time seat availability using backend database and WebSockets
- Coupon codes and offers system

12. Sample Code

File Structure:



```

1 <div class="movie-detail-container">
2   <!-- Loading State -->
3   <div *ngif="loading" class="loading-state">
4     <mat-spinner diameter="50"></mat-spinner>
5     <p>loading movie details...</p>
6   </div>
7
8   <!-- Error State -->
9   <div *ngif="error && !loading" class="error-state">
10     <mat-icon class="error-icon">error_outline</mat-icon>
11     <h3>Movie Not Found</h3>
12     <p>The movie you're looking for doesn't exist or has been removed.</p>
13     <button mat-raised-button color="primary" routerLink="/movies">
14       <mat-icon>arrow_back</mat-icon>
15       Back to Movies
16     </button>
17   </div>
18
19   <!-- Movie Content -->
20   <div *ngif="movie && !loading" class="movie-content">
21     <!-- ... other movie content ... -->
22
23     <mat-tab-group class="movie-tabs">
24       <!-- Description Tab -->
25       <mat-tab label="Description">
26         <div class="tab-content">
27           <h3>Storyline</h3>
28           <p class="description">{{movie.description}}</p>
29         </div>
30       </mat-tab>
31
32       <!-- Showtimes Tab -->
33       <mat-tab label="Showtimes">
34         <div class="tab-content">
35           <!-- ... date and theatre selection ... -->
36         </div>
37     </mat-tab-group>
38   </div>
39 </div>

```

```

20 <div *ngif="movie && !loading" class="movie-content">
21   <mat-tab-group class="movie-tabs">
22     <mat-tab label="Showtimes">
23       <div class="tab-content">
24
25         <!-- Details Section -->
26         <div class="details-section">
27           <div class="screen">Screen {{showtime.screenNumber}}</div>
28           <div class="format">{{showtime.format}}</div>
29           <div class="seats">
30             <mat-icon>event_seat</mat-icon>
31             {{showtime.availableSeats}}/{{showtime.totalSeats}} seats
32           </div>
33         </div>
34
35         <!-- Price Section - CORRECTED -->
36         <div class="price-section">
37           <div class="price">${{showtime.price}}</div>
38           <button
39             mat-raised-button
40             color="accent"
41             (click)="bookShowtime(showtime.id)"
42             >Book Now
43           </button>
44         </div>
45       </div>
46     </mat-tab>
47   </mat-tab-group>
48 </div>

```



```

File Edit Selection View Go Run Terminal Help
MovieBooking

EXPLORER
MOVIEBOOKING
  movie-ticket-system
  angular
  vscode
  node_modules
  public
  src
  app
    components
      booking-summary
      movie-detail
        movie-list.component.html
        movie-detail.component.scss
        movie-detail.component.ts
      movie-list
        movie-list.component.html
        movie-list.component.scss
        movie-list.component.ts
      navbar
      seat-selection
      showtime-list
      theatre-management
      user-bookings
      models
      pages
        home
          home.component.html
          home.component.scss
          home.component.ts
        my-bookings
          my-bookings.component.html
          my-bookings.component.ts
      OUTLINE
      TIMELINE

n src > app > components > movie-list > movie-list.component.html > div.movie-list-container > mat-card.filters-card > div.filters-container > mat-form-field.search-field > input

1 <div class="movie-list-container">
2 <!-- Header -->
3 <div class="page-header">
4 <h1>Movies</h1>
5 <p>Browse our collection of movies and book your tickets</p>
6 </div>
7
8 <!-- Filters Section -->
9 <mat-card class="filters-card">
10 <div class="filters-container">
11 <!-- Search -->
12 <mat-form-field appearance="outline" class="search-field">
13 <mat-label>Search Movies</mat-label>
14 <input
15 | matInput
16 | [(ngModel)]="searchQuery"
17 | (ngModelChange)="filterMovies()"
18 | placeholder="Search by title or description">
19 <mat-icon matSuffix>search</mat-icon>
20 </mat-form-field>
21
22 <!-- Genre Filter -->
23 <mat-form-field appearance="outline">
24 <mat-label>Genre</mat-label>
25 <mat-select [(value)]="selectedGenre" (selectionChange)="filterMovies()">
26 <mat-option value="">All Genres</mat-option>
27 <mat-option *ngFor="let genre of genres" [value]="genre">
28 | {{genre}}
29 </mat-option>
30 </mat-select>
31 </mat-form-field>
32
33 <!-- Language Filter -->
34 <mat-form-field appearance="outline">
35 <mat-label>Language</mat-label>
36 <mat-select [(value)]="selectedLanguage" (selectionChange)="filterMovies()">

```

```

File Edit Selection View Go Run Terminal Help
MovieBooking

EXPLORER
MOVIEBOOKING
  movie-ticket-system
  angular
  vscode
  node_modules
  public
  src
  app
    components
      booking-summary
      movie-detail
        movie-list.component.html
        movie-detail.component.scss
        movie-detail.component.ts
      movie-list
        movie-list.component.html
        movie-list.component.scss
        movie-list.component.ts
      navbar
      seat-selection
      showtime-list
      theatre-management
      user-bookings
      models
      pages
        home
          home.component.html
          home.component.scss
          home.component.ts
        my-bookings
          my-bookings.component.html
          my-bookings.component.ts
      OUTLINE
      TIMELINE

n src > app > components > movie-list > movie-list.component.html > div.movie-list-container > mat-card.filters-card > div.filters-container > mat-form-field.search-field > input

72 <div class="movies-grid" *ngIf="loading && paginatedMovies.length > 0">
73 <mat-card *ngFor="let movie of paginatedMovies" class="movie-card">
134 <mat-card-actions>
148 </mat-card-actions>
149 </mat-card>
150 </div>
151
152 <!-- No Results -->
153 <div *ngIf="loading && filteredMovies.length === 0" class="no-results">
154 <mat-icon class="no-results-icon">theaters</mat-icon>
155 <h3>No movies found</h3>
156 <p>Try changing your search criteria or filters</p>
157 <button mat-raised-button color="primary" (click)="clearFilters()">
158 | Clear All Filters
159 </button>
160 </div>
161
162 <!-- Pagination -->
163 <div *ngIf="filteredMovies.length > pageSize" class="pagination-container">
164 <mat-paginator
165 | [length]="filteredMovies.length"
166 | [pageSize]="pageSize"
167 | [pageSizeOptions]="pageSizeOptions"
168 | [pageIndex]="pageIndex"
169 | (page)="onPageChange($event)"
170 | aria-label="Select page">
171 </mat-paginator>
172 </div>
173
174 <!-- Loading State -->
175 <div *ngIf="loading" class="loading-state">
176 <mat-spinner diameter="50"></mat-spinner>
177 <p>Loading movies...</p>
178 </div>
179
180 </div>

```

```

1  .movie-list-container {
2    max-width: 1200px;
3    margin: 0 auto;
4    padding: 20px;
5
6    .page-header {
7      text-align: center;
8      margin-bottom: 40px;
9
10     h1 {
11       font-size: 2.5rem;
12       color: #333;
13       margin-bottom: 10px;
14     }
15
16     p {
17       color: #666;
18       font-size: 1.1rem;
19     }
20
21     .filters-card {
22       margin-bottom: 30px;
23       border-radius: 12px;
24       overflow: hidden;
25
26       .filters-container {
27         display: grid;
28         grid-template-columns: repeat(auto-fit, minmax(200px, 1fr));
29         gap: 20px;
30         padding: 20px;
31
32         .search-field {
33           grid-column: 1 / -1;
34         }
35
36         mat-form-field {
37

```

```

283  @media (max-width: 768px) {
284    .movie-list-container {
285      .movie-card {
286        .card-image {
287          height: 280px;
288        }
289
290        mat-card-content {
291          padding: 15px;
292
293          .movie-title {
294            font-size: 1.1rem;
295          }
296
297          mat-card-actions {
298            padding: 0 15px 15px;
299            flex-direction: column;
300            gap: 10px;
301          }
302        }
303      }
304    }
305
306    @media (max-width: 480px) {
307      .movie-list-container {
308        .movies-grid {
309          grid-template-columns: 1fr;
310        }
311
312        .movie-card {
313          .card-image {
314            height: 280px;
315          }
316        }
317      }
318    }
319  }
320
321  }
322
323  }
324
325  }
326
327  }
328
329  }
330
331  }
332
333  }
334
335  }
336
337  }
338
339  }
340
341  }
342

```



```

File Edit Selection View Go Run Terminal Help
Q MovieBooking

EXPLORER
MOVIEBOOKING
  movie-ticket-system
  .angular
  .vscode
  node_modules
  public
  src
  app
    components
      booking-summary
        booking-summary.component.html
        booking-summary.component.scss
        TS booking-summary.component.ts
      movie-detail
      movie-list
      navbar
      seat-selection
      showtime-list
      theatre-management
      user-bookings
    models
    pages
      home
        home.component.html
        home.component.scss
        TS home.component.ts
      my-bookings
        my-bookings.component.html
        my-bookings.component.scss
        TS my-bookings.component.ts
    services

OUTLINE
TIMELINE

TS booking-summary.component.ts
movie-ticket-system > src > app > components > booking-summary > TS booking-summary.component.ts > BookingSummaryComponent
1 import { Component, OnInit } from '@angular/core';
2 import { CommonModule } from '@angular/common';
3 import { Router, RouterModule } from '@angular/router';
4 import { MatCardModule } from '@angular/material/card';
5 import { MatButtonModule } from '@angular/material/button';
6 import { MatIconModule } from '@angular/material/icon';
7 import { MatDividerModule } from '@angular/material/divider';
8 import { MatSnackBar } from '@angular/material/snack-bar';
9 import { BookingService } from '../services/booking.service';
10 import { Seat } from '../models/seat.model';
11 import { Showtime } from '../models/showtime.model';
12
13 @Component({
14   selector: 'app-booking-summary',
15   standalone: true,
16   imports: [
17     CommonModule, RouterModule, MatCardModule,
18     MatButtonModule, MatIconModule, MatDividerModule
19   ],
20   templateUrl: './booking-summary.component.html',
21   styleUrls: ['./booking-summary.component.scss']
22 })
23 export class BookingSummaryComponent implements OnInit {
24   selectedSeats: Seat[] = [];
25   showtime: Showtime | null = null;
26   totalPrice = 0;
27   bookingInProgress = false;
28   bookingComplete = false;
29   bookingCode = '';
30
31   constructor(
32     private bookingService: BookingService,
33     private router: Router,
34     private snackBar: MatSnackBar
35   ) {}
36
37   ngOnInit(): void {

```

```

File Edit Selection View Go Run Terminal Help
Q MovieBooking

EXPLORER
MOVIEBOOKING
  movie-ticket-system
  .angular
  .vscode
  node_modules
  public
  src
  app
    components
      booking-summary
        booking-summary.component.html
        booking-summary.component.scss
        TS booking-summary.component.ts
      movie-detail
      movie-list
      navbar
      seat-selection
      showtime-list
      theatre-management
      user-bookings
    models
    pages
      home
        home.component.html
        home.component.scss
        TS home.component.ts
      my-bookings
        my-bookings.component.html
        my-bookings.component.scss
        TS my-bookings.component.ts
    services

OUTLINE
TIMELINE

TS booking-summary.component.ts
movie-ticket-system > src > app > components > booking-summary > TS booking-summary.component.ts > BookingSummaryComponent
23 export class BookingSummaryComponent implements OnInit {
54   confirmBooking(): void {
55     this.bookingInProgress = true;
56
57     setTimeout(() => {
58       this.bookingComplete = true;
59       this.bookingCode = this.generateBookingCode();
60       this.bookingInProgress = false;
61
62       this.snackBar.open('Booking confirmed successfully!', 'OK', { duration: 5000 });
63     }, 2000);
64   }
65
66   generateBookingCode(): string {
67     const chars = 'ABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789';
68     let code = '';
69     for (let i = 0; i < 8; i++) {
70       code += chars.charAt(Math.floor(Math.random() * chars.length));
71     }
72     return code;
73   }
74
75   getSeatTypes(): string {
76     const types = [...new Set(this.selectedSeats.map(seat => seat.type))];
77     return types.map(t => t.charAt(0).toUpperCase() + t.slice(1)).join(', ');
78   }
79
80   formatTime(time: string): string {
81     return new Date(time).toLocaleTimeString('en-US', {
82       hour: '2-digit', minute: '2-digit'
83     });
84   }
85
86   printTicket(): void {
87     window.print();
88   }
89 }

```

```

File Edit Selection View Go Run Terminal Help
movie-ticket-system > src > app > components > movie-list > movie-list.component.ts > MovieListComponent > constructor

1 import { Component, OnInit } from '@angular/core';
2 import { CommonModule } from '@angular/common';
3 import { Router, RouterModule } from '@angular/router';
4 import { FormsModule } from '@angular/forms';
5 import { MatCardModule } from '@angular/material/card';
6 import { MatButtonModule } from '@angular/material/button';
7 import { MatIconModule } from '@angular/material/icon';
8 import { MatInputModule } from '@angular/material/input';
9 import { MatSelectModule } from '@angular/material/select';
10 import { MatProgressSpinnerModule } from '@angular/material/progress-spinner';
11 import { MatPaginatorModule, PageEvent } from '@angular/material/paginator';
12 import { MatChipsModule } from '@angular/material/chips';
13 import { MovieService } from '../services/movie.service';
14 import { Movie } from '../models/movie.model';
15
16 @Component({
17   selector: 'app-movie-list',
18   standalone: true,
19   imports: [
20     CommonModule,
21     RouterModule,
22     FormsModule,
23     MatCardModule,
24     MatButtonModule,
25     MatIconModule,
26     MatInputModule,
27     MatSelectModule,
28     MatProgressSpinnerModule,
29     MatPaginatorModule,
30     MatChipsModule
31   ],
32   templateUrl: './movie-list.component.html',
33   styleUrls: ['./movie-list.component.scss']
34 })
35 export class MovieListComponent implements OnInit {
36   movies: Movie[] = [];
37   filteredMovies: Movie[] = [];

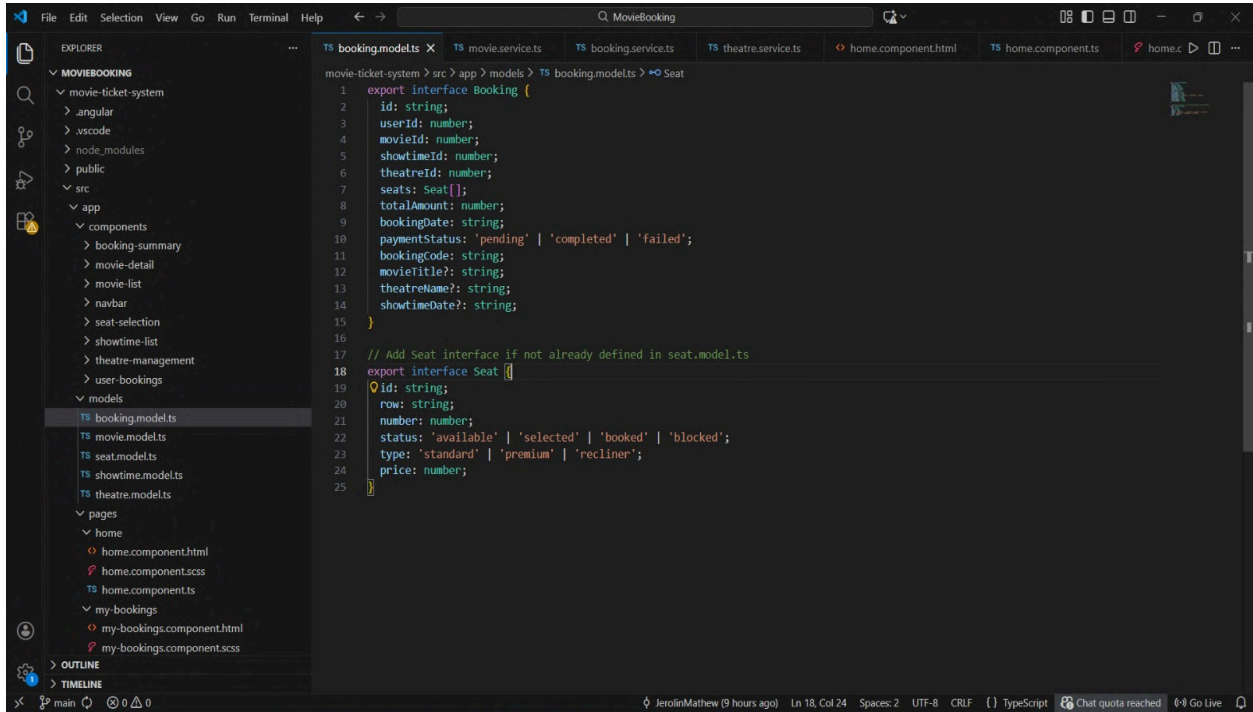
```

```

File Edit Selection View Go Run Terminal Help
movie-ticket-system > src > app > components > movie-list > movie-list.component.ts > MovieListComponent > constructor

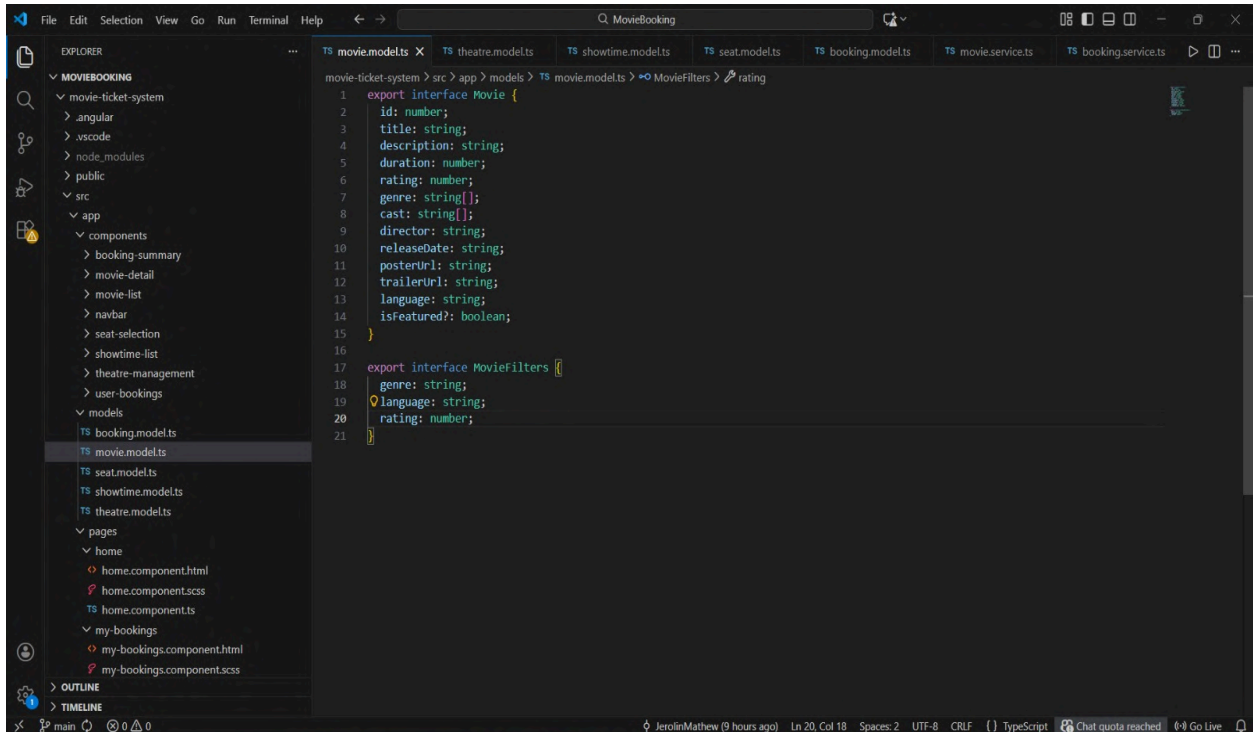
35 export class MovieListComponent implements OnInit {
98 }
99
100 clearFilters(): void {
101   this.searchQuery = '';
102   this.selectedGenre = '';
103   this.selectedLanguage = '';
104   this.minRating = 0;
105   this.filterMovies();
106 }
107
108 onPageChange(event: PageEvent): void {
109   this.pageIndex = event.pageIndex;
110   this.pageSize = event.pageSize;
111 }
112
113 get paginatedMovies(): Movie[] {
114   const startIndex = this.pageIndex * this.pageSize;
115   return this.filteredMovies.slice(startIndex, startIndex + this.pageSize);
116 }
117
118 viewMovieDetails(movieId: number): void {
119   this.router.navigate(['/movie', movieId]);
120 }
121
122 bookTickets(movieId: number): void {
123   this.router.navigate(['/movie', movieId, 'showtimes']);
124 }
125
126 formatDuration(duration: number): string {
127   const hours = Math.floor(duration / 60);
128   const minutes = duration % 60;
129   return `${hours}h ${minutes}m`;
130 }
131 }

```



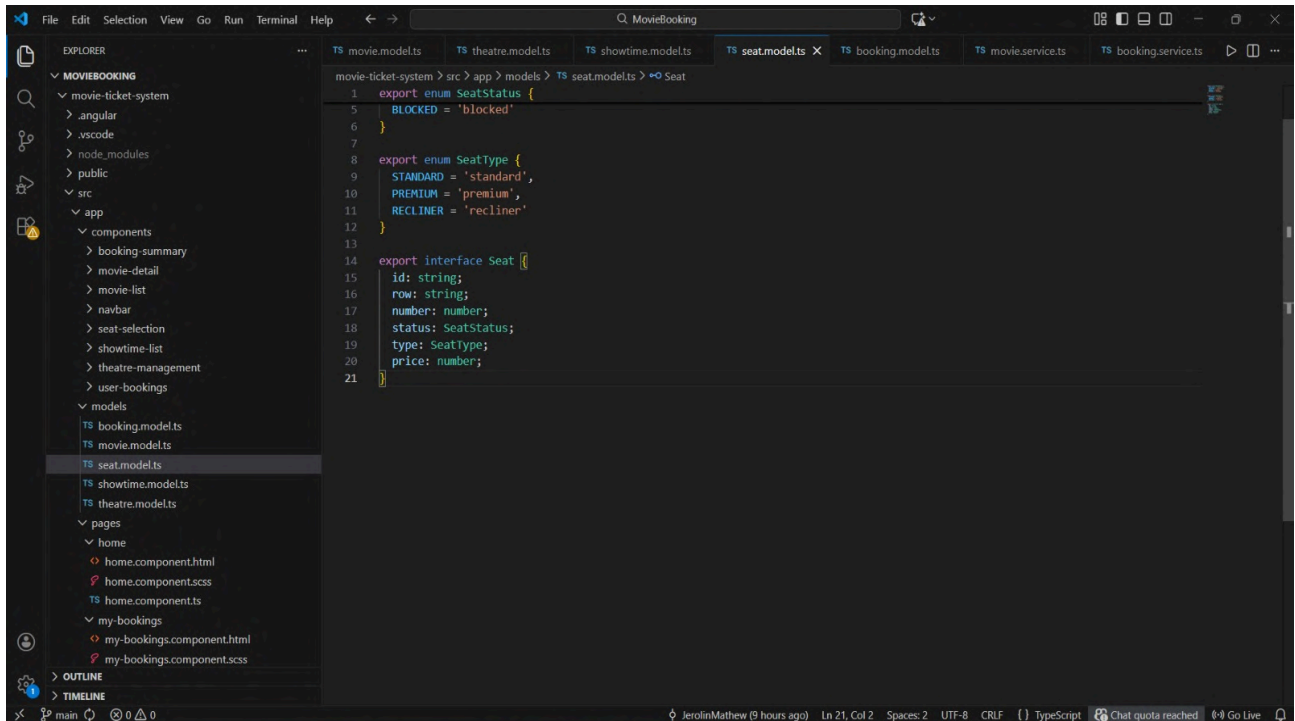
The image shows a VS Code editor window with the Explorer sidebar on the left. The Explorer sidebar shows a project structure for 'MOVIEBOOKING' with folders like 'movie-ticket-system', 'src', 'app', 'components', 'models', 'pages', and 'my-bookings'. The 'models' folder is expanded, showing files like 'booking.models.ts', 'movie.models.ts', 'seat.models.ts', 'showtime.models.ts', and 'theatre.models.ts'. The 'booking.models.ts' file is selected and its content is displayed in the main editor area. The code defines an interface 'Booking' and a 'Seat' interface, with a comment indicating that the 'Seat' interface is added if not already defined in 'seat.models.ts'.

```
1 export interface Booking {
2   id: string;
3   userId: number;
4   movieId: number;
5   showtimeId: number;
6   theatreId: number;
7   seats: Seat[];
8   totalAmount: number;
9   bookingDate: string;
10  paymentStatus: 'pending' | 'completed' | 'failed';
11  bookingCode: string;
12  movieTitle?: string;
13  theatreName?: string;
14  showtimeDate?: string;
15 }
16
17 // Add Seat interface if not already defined in seat.models.ts
18 export interface Seat {
19   id: string;
20   row: string;
21   number: number;
22   status: 'available' | 'selected' | 'booked' | 'blocked';
23   type: 'standard' | 'premium' | 'recliner';
24   price: number;
25 }
```



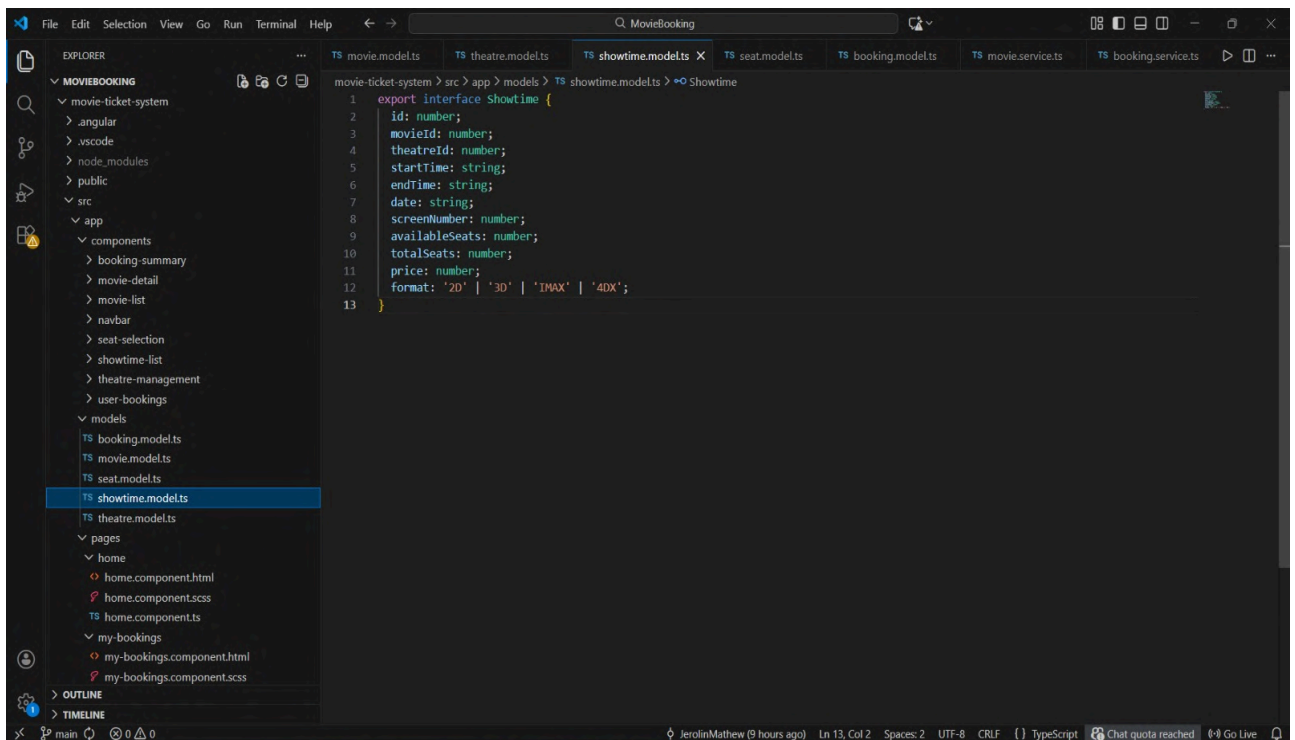
The image shows a VS Code editor window with the Explorer sidebar on the left. The Explorer sidebar shows the same project structure as the previous image. The 'models' folder is expanded, and the 'movie.models.ts' file is selected. The content of 'movie.models.ts' is displayed in the main editor area. The code defines an interface 'Movie' and an interface 'Moviefilters'.

```
1 export interface Movie {
2   id: number;
3   title: string;
4   description: string;
5   duration: number;
6   rating: number;
7   genre: string[];
8   cast: string[];
9   director: string;
10  releaseDate: string;
11  posterUrl: string;
12  trailerUrl: string;
13  language: string;
14  isFeatured?: boolean;
15 }
16
17 export interface Moviefilters {
18   genre: string;
19   language: string;
20   rating: number;
21 }
```



The image shows a VS Code editor window with the Explorer sidebar on the left. The Explorer sidebar is expanded to show the 'models' folder, which contains several TypeScript files: 'booking.models.ts', 'movie.models.ts', 'seat.models.ts' (selected), 'showtime.models.ts', and 'theatre.models.ts'. The main editor area displays the content of 'seat.models.ts'. The code defines two enums, 'SeatStatus' and 'SeatType', and an interface 'Seat'. The 'SeatStatus' enum has a value 'BLOCKED'. The 'SeatType' enum has values 'STANDARD', 'PREMIUM', and 'RECLINER'. The 'Seat' interface has properties: 'id' (string), 'row' (string), 'number' (number), 'status' (SeatStatus), 'type' (SeatType), and 'price' (number).

```
1 export enum SeatStatus {
2     BLOCKED = 'blocked'
3 }
4
5 export enum SeatType {
6     STANDARD = 'standard',
7     PREMIUM = 'premium',
8     RECLINER = 'recliner'
9 }
10
11 export interface Seat {
12     id: string;
13     row: string;
14     number: number;
15     status: SeatStatus;
16     type: SeatType;
17     price: number;
18 }
```



The image shows a VS Code editor window with the Explorer sidebar on the left. The Explorer sidebar is expanded to show the 'models' folder, which contains several TypeScript files: 'booking.model.ts', 'movie.models.ts', 'seat.models.ts', 'showtime.models.ts' (selected), and 'theatre.model.ts'. The main editor area displays the content of 'showtime.models.ts'. The code defines an interface 'Showtime' with properties: 'id' (number), 'movieId' (number), 'theatreId' (number), 'startTime' (string), 'endTime' (string), 'date' (string), 'screenNumber' (number), 'availableSeats' (number), 'totalSeats' (number), 'price' (number), and 'format' (string). The 'format' property has a union type: '2D' | '3D' | 'IMAX' | '4DX'.

```
1 export interface Showtime {
2     id: number;
3     movieId: number;
4     theatreId: number;
5     startTime: string;
6     endTime: string;
7     date: string;
8     screenNumber: number;
9     availableSeats: number;
10    totalSeats: number;
11    price: number;
12    format: '2D' | '3D' | 'IMAX' | '4DX';
13 }
```


File Edit Selection View Go Run Terminal Help

MovieBooking

EXPLORER

- MOVIEBOOKING
 - movie-ticket-system
 - src
 - app
 - components
 - booking-summary
 - movie-detail
 - movie-list
 - navbar
 - seat-selection
 - showtime-list
 - theatre-management
 - user-bookings
 - models
 - pages
 - home
 - home.component.html
 - home.component.scss
 - home.component.ts
 - my-bookings
 - my-bookings.component.html
 - my-bookings.component.scss
 - my-bookings.component.ts
 - services
 - app.component.html
 - app.component.scss
 - app.component.ts
 - app.config.ts
 - app.routes.ts
 - index.html
 - main.ts

movie-ticket-system > src > app > pages > home > home.component.html

```

1 <div class="home-container">
2   <section class="quick-booking-section">
3     <mat-card class="booking-card">
4       <mat-card-content>
5         <div class="steps-container">
6           <div class="step">
7             <div class="step-icon">
8               <mat-icon>schedule</mat-icon>
9             </div>
10            <h4>Select Showtime</h4>
11            <p>Pick date, time and theatre</p>
12          </div>
13          <div class="step">
14            <div class="step-icon">
15              <mat-icon>event_seat</mat-icon>
16            </div>
17            <h4>Book Seats</h4>
18            <p>Choose your preferred seats and confirm</p>
19          </div>
20        </div>
21      </mat-card-content>
22    </mat-card>
23  </section>
24  <!-- Loading State -->
25  <div *ngIf="loading" class="loading-state">
26    <mat-spinner diameter="50"></mat-spinner>
27    <p>Loading movies...</p>
28  </div>
29 </div>

```

Ln 13, Col 1 Spaces: 2 UTF-8 CRLF {} HTML Chat quota reached Go Live

File Edit Selection View Go Run Terminal Help

MovieBooking

EXPLORER

- MOVIEBOOKING
 - movie-ticket-system
 - src
 - app
 - components
 - booking-summary
 - movie-detail
 - movie-list
 - navbar
 - seat-selection
 - showtime-list
 - theatre-management
 - user-bookings
 - models
 - pages
 - home
 - home.component.html
 - home.component.scss
 - home.component.ts
 - my-bookings
 - my-bookings.component.html
 - my-bookings.component.scss
 - my-bookings.component.ts
 - services
 - app.component.html
 - app.component.scss
 - app.component.ts
 - app.config.ts
 - app.routes.ts
 - index.html
 - main.ts

movie-ticket-system > src > app > pages > home > home.component.scss

```

1 .home-container {
2   padding: 0;
3   max-width: 100%;
4   overflow-x: hidden;
5 }
6
7 // Hero Section
8 .hero-section {
9   position: relative;
10  height: 500px;
11  background: linear-gradient(to top right, transparent 49%, #000000 49%, #000000 51%, #ffffff 51%),
12    url('https://images.unsplash.com/photo-148959089516-9827b6d1cf13?ixlib=rb-1.2.1&auto=format&fit=crop&w=1950&q=80');
13  background-size: cover;
14  background-position: center;
15  display: flex;
16  align-items: center;
17  justify-content: center;
18  text-align: center;
19  color: white;
20  margin-bottom: 60px;
21 }
22
23 .hero-content {
24   z-index: 2;
25   position: relative;
26   padding: 0 20px;
27 }
28
29 .hero-title {
30   font-size: 2.5rem;
31   font-weight: bold;
32   margin-bottom: 20px;
33   text-shadow: 2px 2px 4px #000000;
34 }
35
36 .hero-subtitle {
37   font-size: 1.1rem;
38   margin-bottom: 30px;
39   opacity: 0.9;
40 }

```

Ln 493, Col 2 Spaces: 2 UTF-8 CRLF {} SCSS Chat quota reached Go Live

File Edit Selection View Go Run Terminal Help

MovieBooking

EXPLORER

- MOVIEBOOKING
 - movie-ticket-system
 - src
 - app
 - components
 - booking-summary
 - movie-detail
 - movie-list
 - navbar
 - seat-selection
 - showtime-list
 - theatre-management
 - user-bookings
 - models
 - pages
 - home
 - home.component.html
 - home.component.scss
 - home.component.ts**
 - my-bookings
 - my-bookings.component.html
 - my-bookings.component.scss
 - my-bookings.component.ts
 - services
 - app.component.html
 - app.component.scss
 - app.component.ts
 - app.config.ts
 - app.routes.ts
 - index.html
 - main.ts
 - OUTLINE
 - TIMELINE

movie-ticket-system > src > app > pages > home > TS home.component.ts > HomeComponent

```

25 export class HomeComponent implements OnInit {
26   featuredMovies: Movie[] = [];
27   nowShowingMovies: Movie[] = [];
28   comingSoonMovies: Movie[] = [];
29   loading = true;
30
31   constructor(private movieService: MovieService) {}
32
33   ngOnInit(): void {
34     this.loadMovies();
35   }
36
37   loadMovies(): void {
38     this.movieService.getMovies().subscribe({
39       next: (movies: Movie[]) => {
40         this.featuredMovies = movies.filter(movie => movie.isFeatured);
41         this.nowShowingMovies = movies.slice(0, 4);
42         this.comingSoonMovies = movies.slice(4, 8);
43         this.loading = false;
44       },
45       error: (error: any) => {
46         console.error('Error loading movies:', error);
47         this.loading = false;
48       }
49     });
50   }
51
52   getGenresString(genres: string[]): string {
53     return genres.join(', ');
54   }
55
56   formatDuration(duration: number): string {
57     const hours = Math.floor(duration / 60);
58     const minutes = duration % 60;
59     return `${hours}h ${minutes}m`;
60   }
61 }

```

JerolimMathew (9 hours ago) Ln 61, Col 2 Spaces: 2 UTF-8 CRLF () TypeScript Chat quota reached Go Live

File Edit Selection View Go Run Terminal Help

MovieBooking

EXPLORER

- MOVIEBOOKING
 - movie-ticket-system
 - src
 - app
 - components
 - booking-summary
 - movie-detail
 - movie-list
 - navbar
 - seat-selection
 - showtime-list
 - theatre-management
 - user-bookings
 - models
 - pages
 - home
 - home.component.html
 - home.component.scss
 - home.component.ts
 - my-bookings.component.html**
 - my-bookings.component.scss
 - my-bookings.component.ts
 - services
 - app.component.html
 - app.component.scss
 - app.component.ts
 - app.config.ts
 - app.routes.ts
 - index.html
 - main.ts
 - main.ts
 - styles.scss
 - editorconfig
 - main.ts
 - OUTLINE
 - TIMELINE

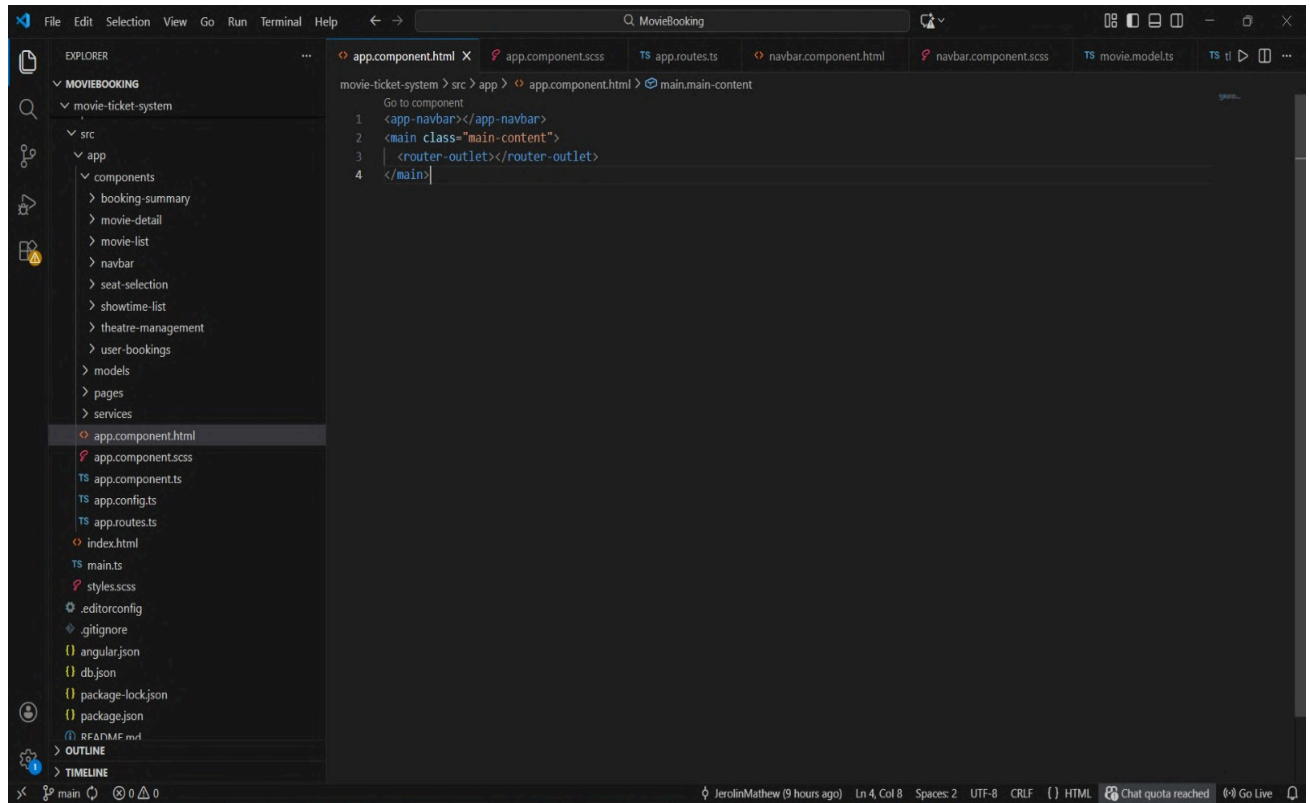
movie-ticket-system > src > app > pages > my-bookings > my-bookings.component.html > div.my-bookings-container

```

1 <div class="my-bookings-container">
2   <!-- Header -->
3   <div class="page-header">
4     <h1>My Bookings</h1>
5     <p>View and manage your movie ticket bookings</p>
6   </div>
7
8   <!-- Loading State -->
9   <div *ngIf="loading" class="loading-state">
10     <mat-spinner diameter="50"></mat-spinner>
11     <p>Loading your bookings...</p>
12   </div>
13
14   <!-- No Bookings -->
15   <div *ngIf="!loading && noBookings" class="no-bookings">
16     <mat-icon class="empty-icon">confirmation_number</mat-icon>
17     <h3>No Bookings Yet</h3>
18     <p>You haven't made any bookings yet.</p>
19     <button mat-raised-button color="primary" routerLink="/movies">
20       <mat-icon>movie</mat-icon>
21       Browse Movies
22     </button>
23   </div>
24
25   <!-- Bookings List -->
26   <div *ngIf="!loading && !noBookings" class="bookings-list">
27     <mat-card *ngFor="let booking of bookings" class="booking-card">
28       <mat-card-header>
29         <div class="booking-header">
30           <div class="booking-code">
31             <mat-icon>confirmation_number</mat-icon>
32             <span>Booking #{{(booking.bookingCode)}}</span>
33           </div>
34           <div class="booking-status" [class]="booking.paymentStatus">
35             {{(booking.paymentStatus | titlecase)}}
36           </div>

```

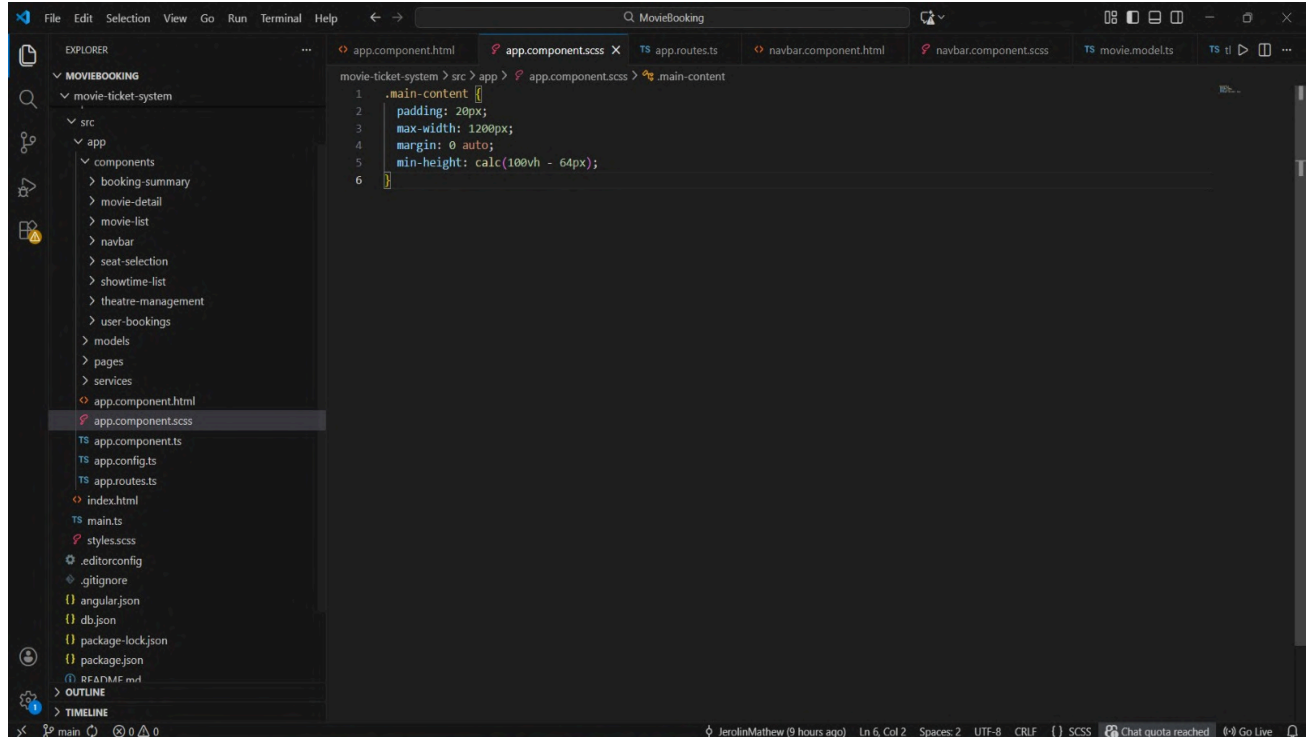
JerolimMathew (9 hours ago) Ln 96, Col 7 Spaces: 2 UTF-8 CRLF () HTML Chat quota reached Go Live



The image shows a Visual Studio Code editor window with the 'movie-ticket-system' project open. The Explorer panel on the left shows the file structure, with 'app.component.html' selected under the 'components' folder. The main editor area displays the content of 'app.component.html' with the following code:

```
1 <app-navbar></app-navbar>
2 <main class="main-content">
3   <router-outlet></router-outlet>
4 </main>
```

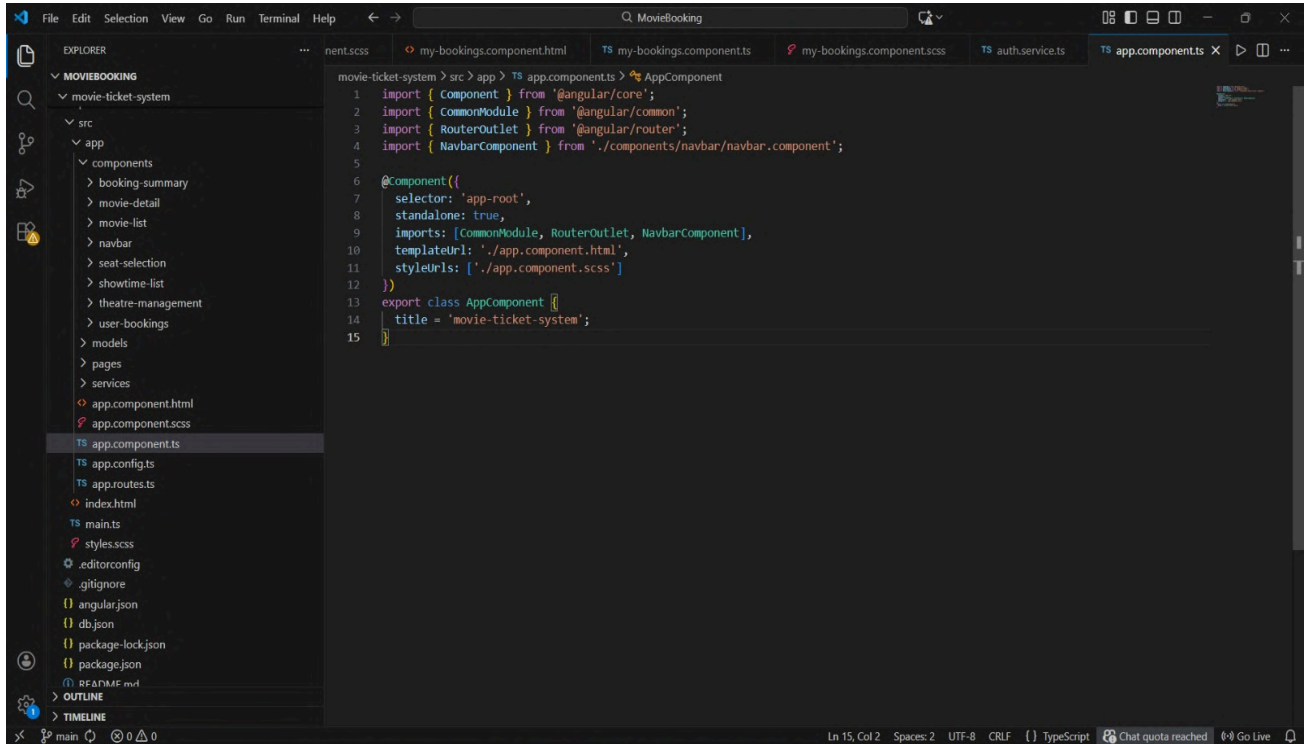
The status bar at the bottom indicates the file is 'app.component.html' at line 4, column 8, with 2 spaces, UTF-8 encoding, and CRLF line endings. The language mode is set to HTML.



The image shows the same Visual Studio Code editor window, but now the 'app.component.scss' file is selected in the Explorer panel. The main editor area displays the content of 'app.component.scss' with the following code:

```
1 .main-content {
2   padding: 20px;
3   max-width: 1200px;
4   margin: 0 auto;
5   min-height: calc(100vh - 64px);
6 }
```

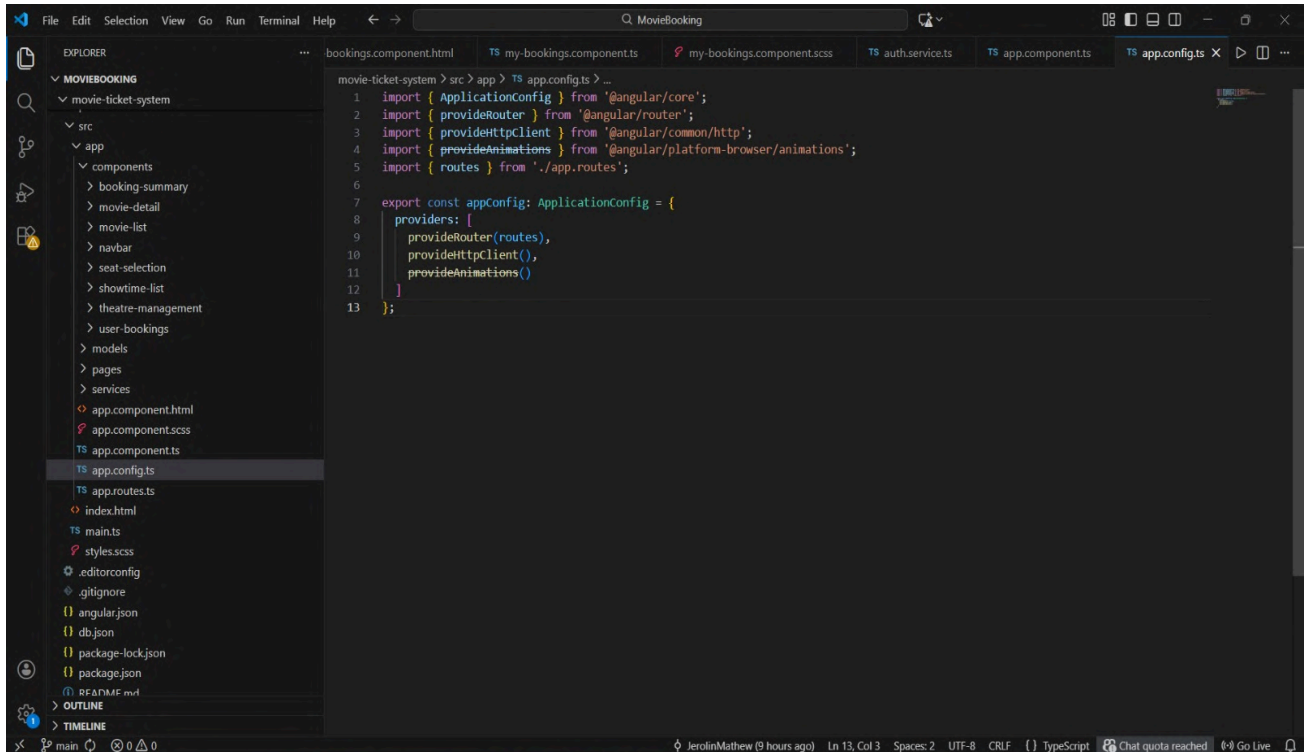
The status bar at the bottom indicates the file is 'app.component.scss' at line 6, column 2, with 2 spaces, UTF-8 encoding, and CRLF line endings. The language mode is set to SCSS.



The image shows a VS Code editor window with the Explorer sidebar on the left. The Explorer sidebar shows the project structure for 'movie-ticket-system'. The 'src' folder is expanded, showing the 'app' folder. The 'app' folder contains several files: 'components', 'booking-summary', 'movie-detail', 'movie-list', 'navbar', 'seat-selection', 'showtime-list', 'theatre-management', 'user-bookings', 'models', 'pages', 'services', 'app.component.html', 'app.component.scss', 'app.component.ts', 'app.config.ts', 'app.routes.ts', 'index.html', 'main.ts', 'styles.scss', '.editorconfig', '.gitignore', 'angular.json', 'db.json', 'package-lock.json', 'package.json', and 'README.md'. The 'app.component.ts' file is selected and its content is displayed in the main editor area.

```

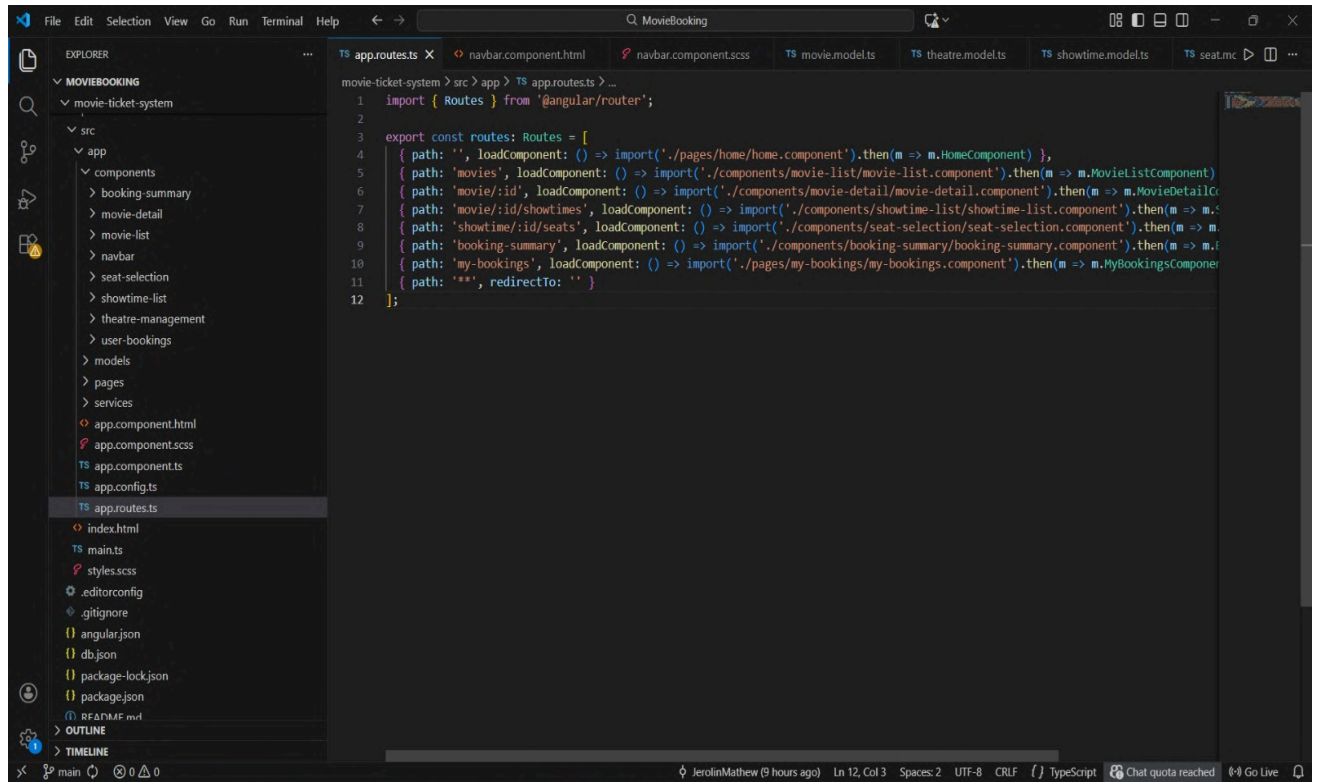
movie-ticket-system > src > app > TS app.component.ts > AppComponent
1  import { Component } from '@angular/core';
2  import { CommonModule } from '@angular/common';
3  import { RouterOutlet } from '@angular/router';
4  import { NavbarComponent } from './components/navbar/navbar.component';
5
6  @Component({
7    selector: 'app-root',
8    standalone: true,
9    imports: [CommonModule, RouterOutlet, NavbarComponent],
10   templateUrl: './app.component.html',
11   styleUrls: ['./app.component.scss']
12 })
13 export class AppComponent {
14   title = 'movie-ticket-system';
15 }
  
```



The image shows a VS Code editor window with the Explorer sidebar on the left. The Explorer sidebar shows the project structure for 'movie-ticket-system'. The 'src' folder is expanded, showing the 'app' folder. The 'app' folder contains several files: 'components', 'booking-summary', 'movie-detail', 'movie-list', 'navbar', 'seat-selection', 'showtime-list', 'theatre-management', 'user-bookings', 'models', 'pages', 'services', 'app.component.html', 'app.component.scss', 'app.component.ts', 'app.config.ts', 'app.routes.ts', 'index.html', 'main.ts', 'styles.scss', '.editorconfig', '.gitignore', 'angular.json', 'db.json', 'package-lock.json', 'package.json', and 'README.md'. The 'app.config.ts' file is selected and its content is displayed in the main editor area.

```

movie-ticket-system > src > app > TS app.config.ts > ...
1  import { ApplicationConfig } from '@angular/core';
2  import { provideRouter } from '@angular/router';
3  import { provideHttpClient } from '@angular/common/http';
4  import { provideAnimations } from '@angular/platform-browser/animations';
5  import { routes } from './app.routes';
6
7  export const appConfig: ApplicationConfig = {
8    providers: [
9      provideRouter(routes),
10     provideHttpClient(),
11     provideAnimations()
12   ]
13 };
  
```

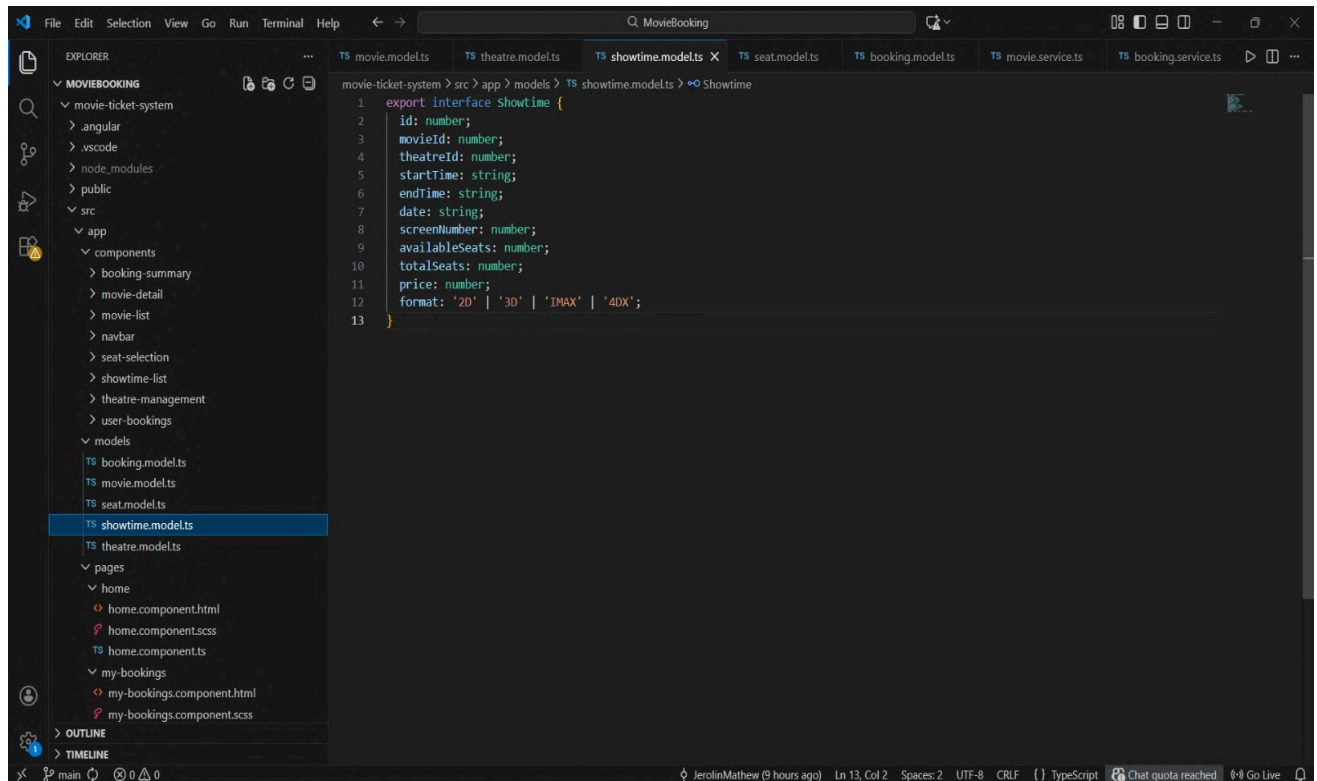


The image shows a VS Code editor window with the file explorer on the left and the editor on the right. The file explorer shows the project structure for 'movie-ticket-system', including 'src', 'app', 'components', 'models', 'pages', 'services', and 'utils'. The 'app.routes.ts' file is selected in the file explorer and is also open in the editor. The editor shows the following code:

```

1 import { Routes } from '@angular/router';
2
3 export const routes: Routes = [
4   { path: '', loadComponent: () => import('./pages/home/home.component').then(m => m.HomeComponent) },
5   { path: 'movies', loadComponent: () => import('./components/movie-list/movie-list.component').then(m => m.MovieListComponent) },
6   { path: 'movie/:id', loadComponent: () => import('./components/movie-detail/movie-detail.component').then(m => m.MovieDetailComponent) },
7   { path: 'movie/:id/showtimes', loadComponent: () => import('./components/showtime-list/showtime-list.component').then(m => m.ShowtimeListComponent) },
8   { path: 'showtime/:id/seats', loadComponent: () => import('./components/seat-selection/seat-selection.component').then(m => m.SeatSelectionComponent) },
9   { path: 'booking-summary', loadComponent: () => import('./components/booking-summary/booking-summary.component').then(m => m.BookingSummaryComponent) },
10  { path: 'my-bookings', loadComponent: () => import('./pages/my-bookings/my-bookings.component').then(m => m.MyBookingsComponent) },
11  { path: '**', redirectTo: '' }
12 ];

```



The image shows a VS Code editor window with the file explorer on the left and the editor on the right. The file explorer shows the project structure for 'movie-ticket-system', including 'src', 'app', 'components', 'models', 'pages', 'services', and 'utils'. The 'showtime.models.ts' file is selected in the file explorer and is also open in the editor. The editor shows the following code:

```

1 export interface Showtime {
2   id: number;
3   movieId: number;
4   theatreId: number;
5   startTime: string;
6   endTime: string;
7   date: string;
8   screenNumber: number;
9   availableSeats: number;
10  totalSeats: number;
11  price: number;
12  format: '2D' | '3D' | 'IMAX' | '4DX';
13 }

```

```

1 <div class="movie-detail-container">
2   <!-- Loading State -->
3   <div *ngIf="loading" class="loading-state">
4     <mat-spinner diameter="50"></mat-spinner>
5     <p>Loading movie details...</p>
6   </div>
7
8   <!-- Error State -->
9   <div *ngIf="error && !loading" class="error-state">
10    <mat-icon class="error-icon">error_outline</mat-icon>
11    <h3>Movie Not Found</h3>
12    <p>The movie you're looking for doesn't exist or has been removed.</p>
13    <button mat-raised-button color="primary" routerLink="/movies">
14      <mat-icon>arrow_back</mat-icon>
15      Back to Movies
16    </button>
17  </div>
18
19  <!-- Movie Content -->
20  <div *ngIf="movie && !loading" class="movie-content">
21    <!-- ... other movie content ... -->
22
23    <mat-tab-group class="movie-tabs">
24      <!-- Description Tab -->
25      <mat-tab label="Description">
26        <div class="tab-content">
27          <h3>Storyline</h3>
28          <p class="description">{{movie.description}}</p>
29        </div>
30      </mat-tab>
31
32      <!-- Showtimes Tab -->
33      <mat-tab label="Showtimes">
34        <div class="tab-content">
35          <!-- ... date and theatre selection ... -->
36        </div>
37      </mat-tab>
38    </mat-tab-group>
39  </div>
40 </div>

```

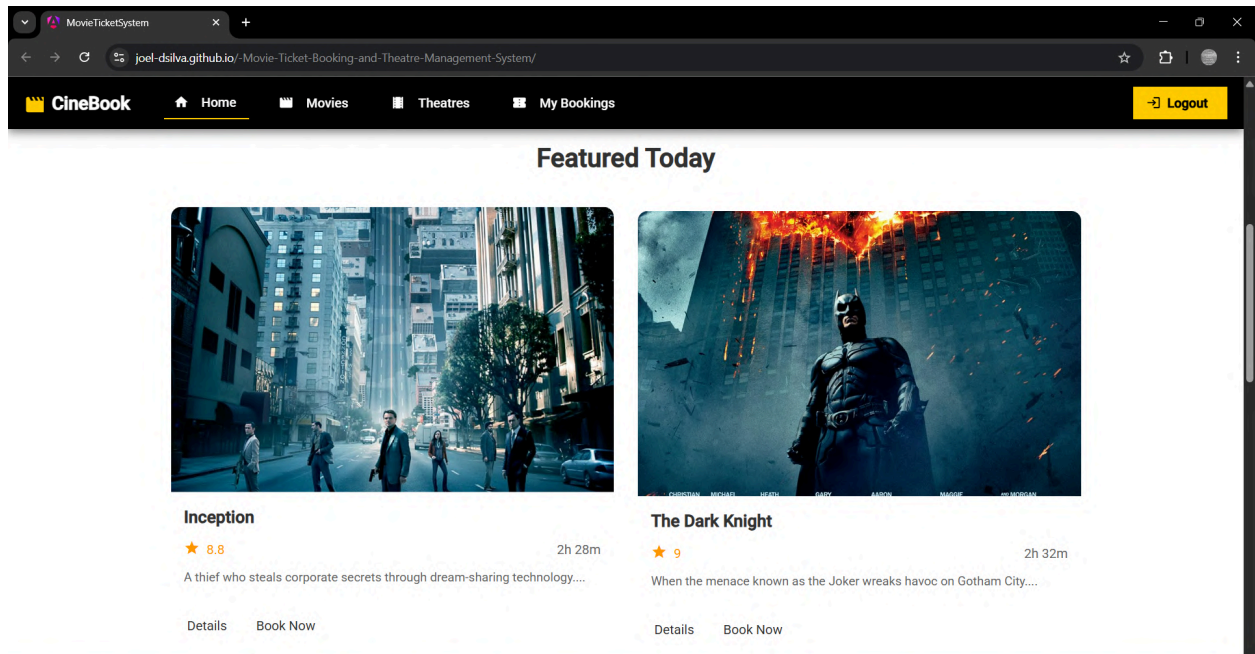
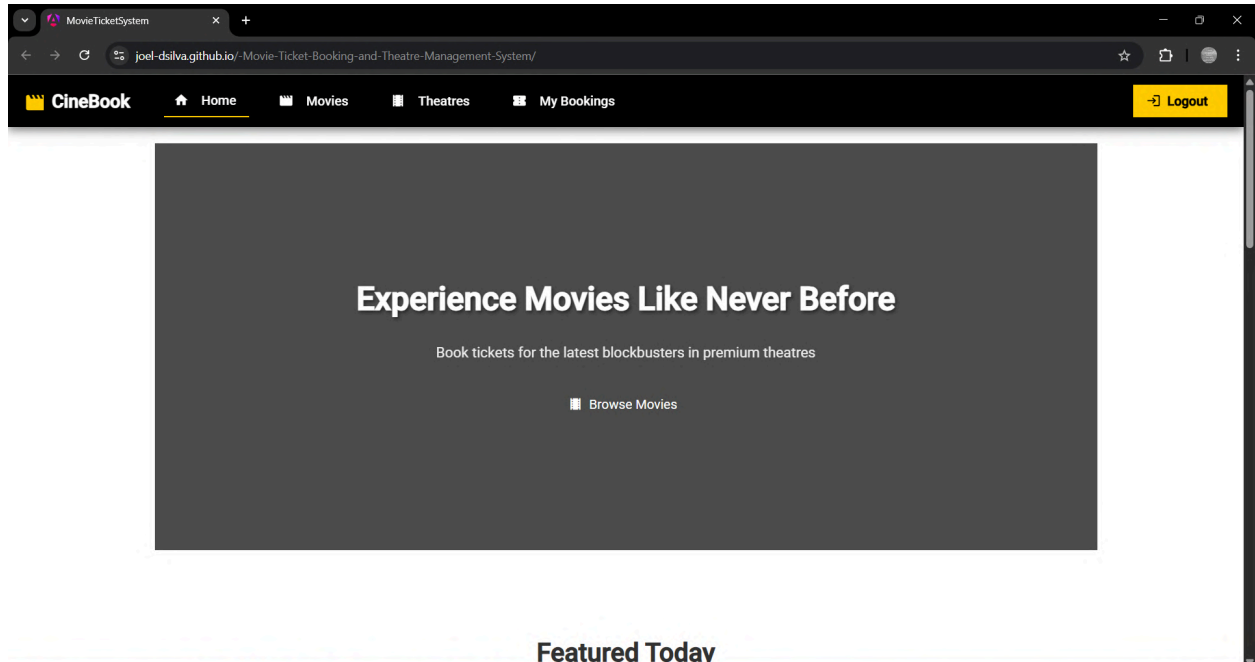
```

20 <div *ngIf="movie && !loading" class="movie-content">
21   <mat-tab-group class="movie-tabs">
22     <mat-tab label="Showtimes">
23       <div class="tab-content">
24
25       <!-- Details Section -->
26       <div class="details-section">
27         <div class="screen">Screen {{showtime.screenNumber}}</div>
28         <div class="format">{{showtime.format}}</div>
29         <div class="seats">
30           <mat-icon>event_seat</mat-icon>
31           {{showtime.availableSeats}}/{{showtime.totalSeats}} seats
32         </div>
33       </div>
34
35       <!-- Price Section - CORRECTED -->
36       <div class="price-section">
37         <div class="price">${{showtime.price}}</div>
38         <button
39           mat-raised-button
40           color="accent"
41           (click)="bookShowtime(showtime.id)">
42           Book Now
43         </button>
44       </div>
45     </mat-tab>
46   </mat-tab-group>
47 </div>

```

13. Screenshots of Final Output

Home:




[Home](#)
[Movies](#)
[Theatres](#)
[My Bookings](#)
[Login](#)

Now Showing

[View All →](#)



Inception
Action, Sci-Fi, Thriller
★ 8.8 2h 28m



The Dark Knight
Action, Crime, Drama
★ 9 2h 32m



Interstellar
Adventure, Drama, Sci-Fi
★ 8.6 2h 49m


[Home](#)
[Movies](#)
[Theatres](#)
[My Bookings](#)
[Login](#)

Inception

Action, Sci-Fi, Thriller

★ 8.8 2h 28m

The Dark Knight

Action, Crime, Drama

★ 9 2h 32m

Interstellar

Adventure, Drama, Sci-Fi

★ 8.6 2h 49m

Quick Booking

Find and book tickets in 3 easy steps



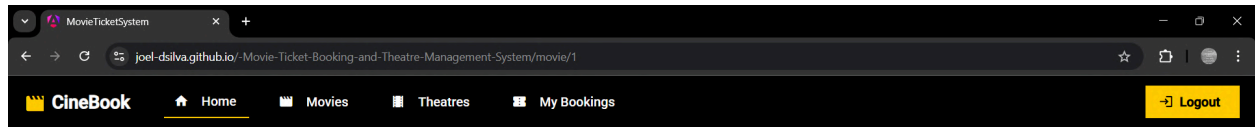
Choose Movie
Browse our collection of latest movies



Select Showtime
Pick date, time and theatre



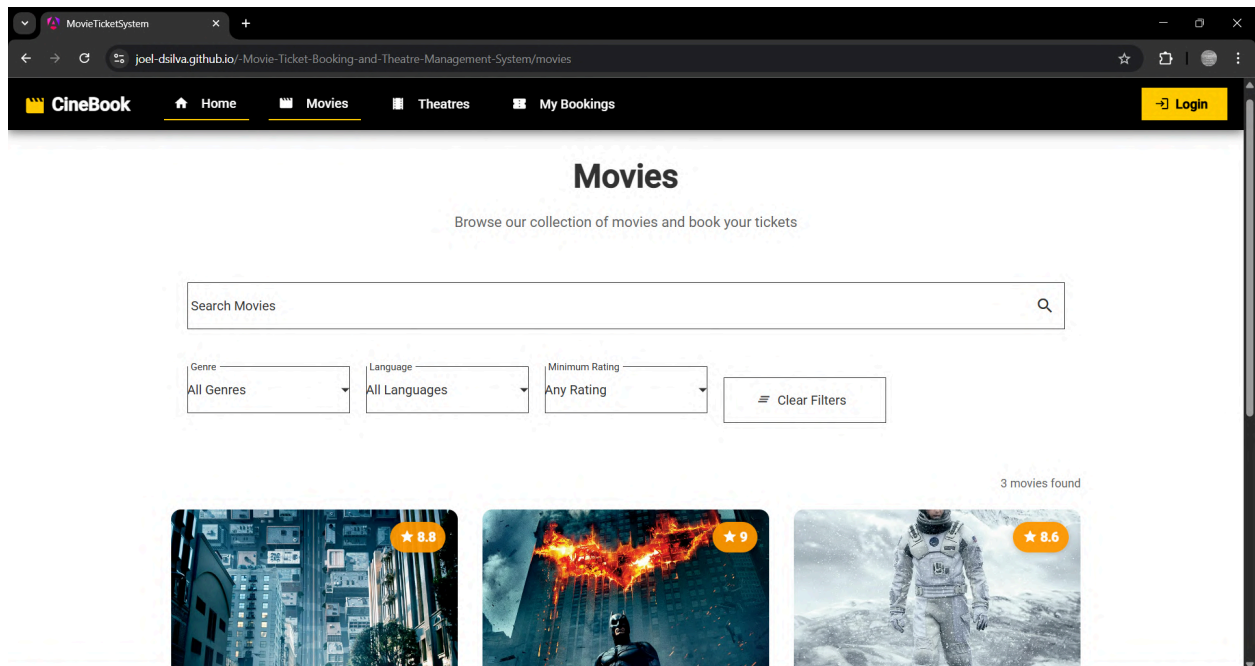
Book Seats
Choose your preferred seats and confirm



Storyline

A thief who steals corporate secrets through dream-sharing technology.

Movies:



MovieTicketSystem

joel-dsilva.github.io/ Movie-Ticket-Booking-and-Theatre-Management-System/movies

CineBook
Home
Movies
Theatres
My Bookings
Login



Inception

2h 28m English

Action Sci-Fi +1 more

Cast: Leonardo DiCaprio, Joseph Gordon-Levitt ...

A thief who steals corporate secrets through dream-sharing technology....

View Details Book Tickets



The Dark Knight

2h 32m English

Action Crime +1 more

Cast: Christian Bale, Heath Ledger ...

When the menace known as the Joker wreaks havoc on Gotham City....

View Details Book Tickets



Interstellar

2h 49m English

Adventure Drama +1 more

Cast: Matthew McConaughey, Anne Hathaway ...

A team of explorers travel through a wormhole in space....

View Details Book Tickets

Theatres:

MovieTicketSystem

joel-dsilva.github.io/ Movie-Ticket-Booking-and-Theatre-Management-System/movie/1/showtimes

CineBook
Home
Movies
Theatres
My Bookings
Logout

Back



Inception

148 min • 8.8 ★ • English

Date
05:30 AM

Theatre
All Theatres

Format
All Formats

All Formats
IMAX

0 Showtimes Available

No showtimes found for selected filters

MovieTicketSystem

joel-dsilva.github.io/ Movie-Ticket-Booking-and-Theatre-Management-System/movie/1/showtimes

CineBook

[Home](#)
[Movies](#)
[Theatres](#)
[My Bookings](#)

Logout

[← Back](#)



Inception

148 min • 8.8 ★ • English

Date
05:30 AM

Theatre
All Theatres

All Theatres ✓
PVR Cinemas

Format
All Formats

0 Showtimes Available



No showtimes found for selected filters

My Bookings:

MovieTicketSystem

joel-dsilva.github.io/ Movie-Ticket-Booking-and-Theatre-Management-System/my-bookings

CineBook

[Home](#)
[Movies](#)
[Theatres](#)
[My Bookings](#)

Logout

My Bookings

View and manage your movie ticket bookings

 Booking #ABC123XYZ

Completed

 Movie
Inception

 Theatre
PVR Cinemas

 Showtime
Wed, Mar 20, 2024, 06:00 PM

 Seats
A5 A6

 Amount
\$25.98

 Print Ticket

 Cancel Booking

 Book Again



11. Conclusion

The Restaurant Menu and Ordering System demonstrates the practical application of Angular in building a real-world web application. This project enhanced our understanding of component-based development, routing, state management, and responsive UI design. It serves as a strong foundation for developing scalable web applications using modern JavaScript frameworks.

12. References

- L&T LMS : <https://learn.lntedutech.com/Landing/MyCourse>
- Angular Documentation – <https://angular.io>